Math 118: Mathematical Methods of Data Theory

Lecture 13: The Simplex Method

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Linear Programming: Example 3

- Suppose we wish to solve **under-determined** linear system $A\boldsymbol{x} = \boldsymbol{b}$ where $A \in \mathbb{R}^{m \times n}$ and $m < n$.
- System has infinitely many solutions, how to choose one?
- One rule: best solution should be the simplest, and so have the **fewest non-zero entries.**:

$$\boldsymbol{x}^* = \underset{\boldsymbol{x} \text{ such that } A\boldsymbol{x}=\boldsymbol{b}}{\arg\min} \{\# \text{ non-zero entries in } \boldsymbol{x}\}$$

- In many cases can find $\boldsymbol{x}^*$ by solving the (easier) Lasso problem:

$$\boldsymbol{x}^* = \underset{\boldsymbol{x}:\ A\boldsymbol{x}=\boldsymbol{b}}{\arg\min} \|\boldsymbol{x}\|_1$$

Linear Programming: Example 3

- Can convert Lasso: $\boldsymbol{x}^* = \underset{\boldsymbol{x}:\ A\boldsymbol{x}=\boldsymbol{b}}{\arg\min} \|\boldsymbol{x}\|_1$ to a Linear Program.
- Write $x_i = [x_i]_+ - [x_i]_- = u_i - v_i$
- **Observe:** $\|\boldsymbol{x}\|_1 = \sum_{i=1}^n u_i + v_i = \mathbf{1}^\top \begin{bmatrix} \boldsymbol{u} \\ \boldsymbol{v} \end{bmatrix}$

$$\boldsymbol{x}^* = \underset{\boldsymbol{x}:\ A\boldsymbol{x}=\boldsymbol{b}}{\arg\min} \|\boldsymbol{x}\|_1 \Leftrightarrow \min \mathbf{1}^\top \begin{bmatrix} \boldsymbol{u} \\ \boldsymbol{v} \end{bmatrix} \text{ subject to: } \begin{bmatrix} \boldsymbol{u} \\ \boldsymbol{v} \end{bmatrix} \geq 0 \text{ and } \begin{bmatrix} A & -A \end{bmatrix} \begin{bmatrix} \boldsymbol{u} \\ \boldsymbol{v} \end{bmatrix} = \boldsymbol{b}$$

- **Recall** Feasible region:

$$\mathcal{X} = \left\{ \begin{bmatrix} \boldsymbol{u} \\ \boldsymbol{v} \end{bmatrix} \in \mathbb{R}^{2n} : \begin{bmatrix} \boldsymbol{u} \\ \boldsymbol{v} \end{bmatrix} \geq 0 \text{ and } \begin{bmatrix} A & -A \end{bmatrix} \begin{bmatrix} \boldsymbol{u} \\ \boldsymbol{v} \end{bmatrix} = \boldsymbol{b} \right\}$$

  is called a polyhedron.

Idea of the Simplex Method

$$\min \boldsymbol{c}^\top \boldsymbol{x} \text{ subject to:} A\boldsymbol{x} = \boldsymbol{b} \text{ and } \boldsymbol{x} \geq 0$$

- If $\boldsymbol{x}^*$ is a solution then it is a corner of $\{\boldsymbol{x} : A\boldsymbol{x} = \boldsymbol{b} \text{ and } \boldsymbol{x} \geq 0\}$.
- **edges** $[\boldsymbol{x}_i, \boldsymbol{x}_j]$ connects the corners $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$.
- **reduced cost:** decrease in cost when traversing an edge $[\boldsymbol{x}_i, \boldsymbol{x}_j]$, *i.e.* $\boldsymbol{c}^\top \boldsymbol{x}_j - \boldsymbol{c}^\top \boldsymbol{x}_i$
- **The Simplex Method:** *(idea)*
  1. Find a corner $\boldsymbol{x}_1$.
  2. For $k = 1 : K_{\max}$
     - 2.1 Examine all corners adjacent to $\boldsymbol{x}_k$. Call them $\boldsymbol{z}_1, \ldots, \boldsymbol{z}_r$.
     - 2.2 Select any $\boldsymbol{z}_j$ such that reduced cost along $[\boldsymbol{x}_i, \boldsymbol{z}_j]$ is negative.
     - 2.3 If no such edge exists, terminate and return $\boldsymbol{x}_* = \boldsymbol{x}_k$. This is the true minimizer.
  3. If $K_{\max}$ reached, return $\boldsymbol{x}_* = \boldsymbol{x}_{K_{\max}}$ an approximate solution.

Identifying Corners

$$\min \boldsymbol{c}^\top \boldsymbol{x} \text{ subject to: } \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \text{ and } \boldsymbol{x} \geq 0$$

- Let $\mathcal{X} = \{\boldsymbol{x} : \ A\boldsymbol{x} = \boldsymbol{b} \text{ and } \boldsymbol{x} \geq 0\}$.
- **Feasible points** are $\boldsymbol{x} \in \mathcal{X}$.
- Assume we have n variables and m (equality) constraints. So $A \in \mathbb{R}^{m \times n}$, with $m \leq n$.
- Shall further assume that rank$(A) = m$.
- Equality constraints $\boldsymbol{a}_i^\top \boldsymbol{x} = b_i$ and $x_i = 0$ correspond to faces of $\mathcal{X}$.
- Corners are feasible points where n faces meet.
- Have m equality constraints so need $x_i = 0$ for $n - m$ different i.
- **Basic Feasible Solution** (aka corner) is a feasible point $\boldsymbol{x} = [\boldsymbol{x}_B, \boldsymbol{0}_{n-m}]^\top$.
- $B \subset \{1, \ldots, n\}$ is called a **basis**.
- Correspondingly: $\boldsymbol{A} = [\boldsymbol{B}, \boldsymbol{D}]$ *(reordering indices if necessary)*.
- **Caution:** Not all basic solutions are feasible. There exist fake corners!

More on Basic Feasible Solutions

- **Basic Feasible Solution** (aka corner) is a feasible point $\boldsymbol{x} = [\boldsymbol{x}_B, \mathbf{0}_{n-m}]^\top$.
- Correspondingly: $\boldsymbol{A} = [\boldsymbol{B}, \boldsymbol{D}]$ *(reordering indices if necessary)*.
- Moreover:

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \Rightarrow [\boldsymbol{B}, \boldsymbol{D}] \begin{bmatrix} \boldsymbol{x}_B \\ \mathbf{0} \end{bmatrix} = \boldsymbol{b} \Rightarrow \boldsymbol{B}\boldsymbol{x} = \boldsymbol{b} \Rightarrow \boldsymbol{x}_B = \boldsymbol{B}^{-1}\boldsymbol{b}$$

- *(and $\boldsymbol{x}_B \geq 0$)*
- B are called the **basic** variables and D are the **non-basic** variables.
- An edge is obtained by **releasing a non-basic variable** x_j from 0.

Edges

- An edge is obtained by **releasing a non-basic variable** x_j from 0.
- Choose $j \in D$, then:

$$\{\boldsymbol{y} \geq 0 : \ \boldsymbol{A}\boldsymbol{y} = \boldsymbol{b}, \ y_i = 0 \ \forall i \in D \setminus \{j\}\} \subset \mathcal{X}$$

  is an edge leaving the BFS $\boldsymbol{x} = [\boldsymbol{x}_B, \mathbf{0}]^\top$.

- Edge is a straight line $\boldsymbol{y} = \boldsymbol{x} + \epsilon\boldsymbol{\delta}_j$ where $\boldsymbol{\delta}_j = \begin{bmatrix} -\boldsymbol{B}^{-1}\boldsymbol{a}_j \\ \mathbf{e}_j \end{bmatrix} = \begin{bmatrix} \boldsymbol{\delta}_{j,B} \\ \boldsymbol{e}_j \end{bmatrix}$
- $\boldsymbol{a}_j$ is j-th *column* of A and $(\boldsymbol{e}_j)_i = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$
- **Check:**

$$A(\boldsymbol{x}+\epsilon\boldsymbol{\delta}_j) = \boldsymbol{A}\boldsymbol{x}+\epsilon \begin{bmatrix} \boldsymbol{B} & \boldsymbol{D} \end{bmatrix} \begin{bmatrix} -\boldsymbol{B}^{-1}\boldsymbol{a}_j \\ \mathbf{e}_j \end{bmatrix} = \boldsymbol{b}+\epsilon\left(-\boldsymbol{a}_j + \boldsymbol{D}\boldsymbol{e}_j\right) = \boldsymbol{b}+\epsilon(0) = \boldsymbol{b}$$

- An **edge** is: $\{\boldsymbol{y} = \boldsymbol{x} + \epsilon\boldsymbol{\delta}_j : \ \boldsymbol{y} \geq 0\}$.

Reduced Cost

- An **edge** is: $\{\boldsymbol{y} = \boldsymbol{x} + \epsilon\boldsymbol{\delta}_j : \ \boldsymbol{y} \geq 0\}$ where $\boldsymbol{\delta}_j = \begin{bmatrix} -\boldsymbol{B}^{-1}\boldsymbol{a}_j \\ \mathbf{e}_j \end{bmatrix}$.
- **Reduced cost** along this edge:

$$\boldsymbol{c}^\top \boldsymbol{y} - \boldsymbol{c}^\top \boldsymbol{x} = \boldsymbol{c}^\top (\boldsymbol{x} + \epsilon\boldsymbol{\delta}_j - \boldsymbol{x}) = \epsilon \boldsymbol{c}^\top \boldsymbol{\delta}_j$$

- $\boldsymbol{c}^\top \boldsymbol{\delta}_j$ is sometimes called the **unit cost** along this edge:

$$\boldsymbol{c}^\top \boldsymbol{\delta}_j = -\boldsymbol{c}_B^\top B^{-1} \boldsymbol{a}_j + c_j$$

- If unit cost is negative, this is a viable edge for simplex method to choose.

Unbounded Edges

- An **edge** is: $\{\boldsymbol{y} = \boldsymbol{x} + \epsilon\boldsymbol{\delta}_j : \ \boldsymbol{y} \geq 0\}$ where $\boldsymbol{\delta}_j = \begin{bmatrix} -\boldsymbol{B}^{-1}\boldsymbol{a}_j \\ \mathbf{e}_j \end{bmatrix}$
- If $-\boldsymbol{B}^{-1}\boldsymbol{a}_j \geq 0$ then $\boldsymbol{y} \geq 0$ for all $\epsilon \geq 0$
- Hence this edge is **unbounded**
- If in addition the unit cost $\boldsymbol{c}^\top \boldsymbol{\delta}_j$ is negative then cost can be made arbitrarily small by proceeding along this edge.
- Hence in this case cost is **unbounded** from below and this Linear Program does not have a solution.

Ends of Edges

- An **edge** is: $\{\boldsymbol{y} = \boldsymbol{x} + \epsilon\boldsymbol{\delta}_j : \ \boldsymbol{y} \geq 0\}$.
- Suppose $-\boldsymbol{B}^{-1}\boldsymbol{a}_j \not\geq 0$ so this edge is **bounded.**
- **End of edge** corresponds to largest ϵ such that $\boldsymbol{y} \geq 0$.
- $\boldsymbol{y} = \begin{bmatrix} \boldsymbol{x}_B - \epsilon B^{-1}\boldsymbol{a}_j \\ \epsilon \boldsymbol{e}_j \end{bmatrix}$ so $\boldsymbol{y} \geq 0$ as long as $\boldsymbol{x}_B - \epsilon B^{-1}\boldsymbol{a}_j \geq 0$.
- Define $\epsilon_{\max} = \min \left\{ \frac{x_i}{(B^{-1}\boldsymbol{a}_j)_i} : i \in B \text{ and } -(B^{-1}\boldsymbol{a}_j)_i < 0 \right\}$
- Define $i^* = \arg\min \left\{ \frac{x_i}{(B^{-1}\boldsymbol{a}_j)_i} i \in B \text{ and } -(B^{-1}\boldsymbol{a}_j)_i < 0 \right\}$
- So, new BFS is $\boldsymbol{x}^{\text{new}} = \begin{bmatrix} \boldsymbol{x}_B - \epsilon_{\max} B^{-1}\boldsymbol{a}_j \\ \epsilon_{\max}\boldsymbol{e}_j \end{bmatrix}$ and $B^{\text{new}} = (B \setminus \{i^*\}) \cup \{j\}$
- Reduced cost along $\{\boldsymbol{y} = \boldsymbol{x} + \epsilon\boldsymbol{\delta}_j : \ \boldsymbol{y} \geq 0\}$ is $\epsilon_{\max}\boldsymbol{c}^\top\boldsymbol{\delta}_j$

The Simplex Method: Main Loop

- **Input** a BFS $\boldsymbol{x}$ with basis B.
- **Compute** unit costs $\bar{c}_j = \boldsymbol{c}^\top \boldsymbol{\delta}_j$ for all $j \in D = B^c$.
- **If** $\bar{c}_j \geq 0$ for all j then $\boldsymbol{x}$ is optimal. **Terminate**.
- **If not** pick any j with $\bar{c}_j < 0$:
  - **If** $\boldsymbol{\delta}_j \geq 0$ then problem is unbounded. **Terminate**.
  - **If not** then compute $\epsilon_{\max}$ and i^*.
  - $\boldsymbol{x} \leftarrow \boldsymbol{x}_B + \epsilon_{\max} \boldsymbol{\delta}_j$ and $B \leftarrow (B \setminus \{i^*\}) \cup \{j\}$

Two Phase Simplex Method

- Finding an initial BFS is hard!
- Solution: solve an easier, initial LP.
- Given a standard form LP, ignore cost function so just have:

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b},\ \boldsymbol{x} \geq 0$$

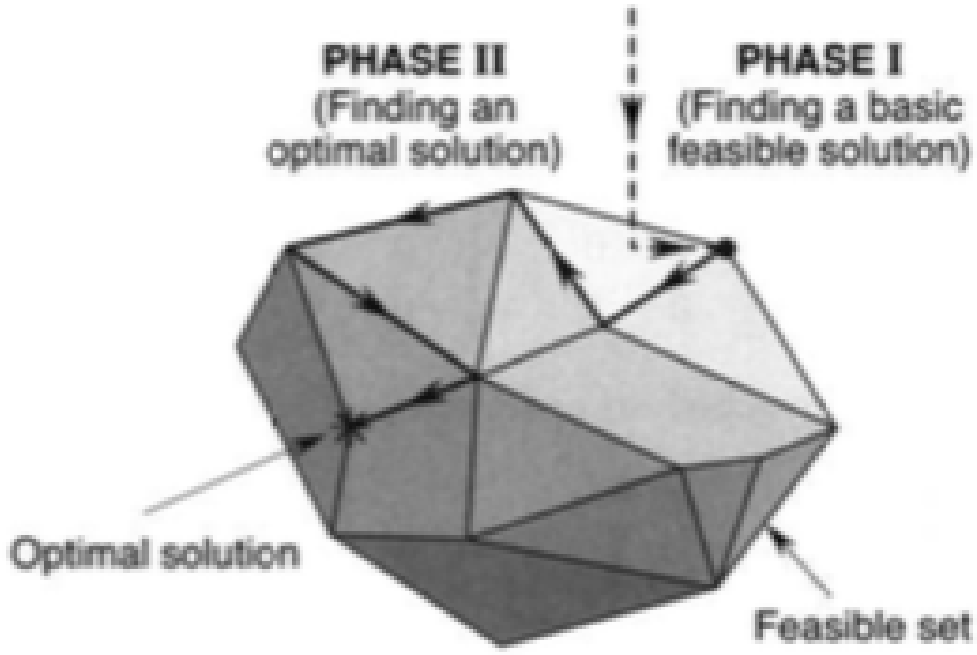
- **Step 1** ensure $\boldsymbol{b} \geq 0$ by $\times(-1)$ to rows with $b_i < 0$.
- **Step 2** add artificial variables $y_1, \ldots, y_m$ and minimize their sum:

$$\min_{\boldsymbol{x},\boldsymbol{y}} y_1 + \ldots + y_m \text{ s.t. } \begin{bmatrix}\boldsymbol{A} & \boldsymbol{I}\end{bmatrix} \begin{bmatrix}\boldsymbol{x} \\ \boldsymbol{y}\end{bmatrix} = \boldsymbol{b}\ \boldsymbol{x}, \boldsymbol{y} \geq 0$$

- Note that this problem has an obvious BFS: $\begin{bmatrix}\boldsymbol{x} & \boldsymbol{y}\end{bmatrix}^\top = \begin{bmatrix}\boldsymbol{0} & \boldsymbol{b}\end{bmatrix}^\top$.
- **Step 3** Solve this problem until $\boldsymbol{y} = \boldsymbol{0}$. Remaining basis is a basis only in $\boldsymbol{x}$.
- **Step 4** Now, solve original problem:

$$\min \boldsymbol{c}^\top \boldsymbol{x} \text{ subject to: } \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \text{ and} \boldsymbol{x} \geq 0$$

Illustration of Two Phase Simplex Method



$n - m = 3$, each 2D face corresponds to some $x_i = 0$

Figure: Taken from "An Introduction to Optimization" by Chong and Zak

Main Theorems for Simplex Method

- If a LP in standard form has an optimal feasible solution, then it has an optimal basic feasible solution. *(That is, if a minimizer exists then there must be a minimizer at a corner)*.
- For 2-phase simplex method, the original LP has a basic feasible solution if and only if the phase 1 LP has an optimal basic feasible solution with 0 objective. *(That is, phase 1 will succesfully find a BFS for use in phase 2, as long as such a BFS exists)*

Example 1

$$\text{maximize } 7x_1 + 6x_2 \text{ subject to: } \begin{array}{c} 2x_1 + x_2 \leq 3 \\ x_1 + 4x_2 \leq 4 \\ x_1, x_2 \geq 0 \end{array}$$

Change to Standard Form:

1. Change from maximize $7x_1 + 6x_2$ to minimize $-7x_1 + -6x_2$. So $\boldsymbol{c}^\top = [-7, -6]$.
2. Slack variables: Introduce x_3 and x_4 with $x_3 = 3 - 2x_1 - x_2$ and $x_4 = 4 - x_1 - 4x_2$
3. Standard form:

$$\text{minimize } -7x_1 - 6x_2 \text{ subject to: } \begin{bmatrix} 2 & 1 & 1 & 0 \\ 1 & 4 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Example 1

$$\text{minimize } \begin{bmatrix} -7 & -6 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \text{ subject to: } \begin{bmatrix} 2 & 1 & 1 & 0 \\ 1 & 4 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Now apply simplex method:

- Find a basis: $B = \{3, 4\}$ works, hence $\boldsymbol{B} = I = \boldsymbol{B}^{-1}$.
- initial **BFS**[1] is $\boldsymbol{x} = [0, 0, 3, 4]^\top$ *(check that this is feasible!)*
- Compute unit costs $\bar{c}_1$ and $\bar{c}_2$:

$$\bar{c}_2 = -\boldsymbol{c}_B^\top \boldsymbol{B}^{-1} \boldsymbol{a}_2 + c_2 = -0 + (-6) = -6$$
$$\bar{c}_1 = -\boldsymbol{c}_B^\top \boldsymbol{B}^{-1} \boldsymbol{a}_1 + c_1 = -0 + (-7) = -7$$

- **Edges**: $\boldsymbol{\delta}_1 = \begin{bmatrix} 1 \\ 0 \\ -\boldsymbol{B}^{-1}\boldsymbol{a}_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -2 \\ -1 \end{bmatrix}$ and $\boldsymbol{\delta}_2 = \begin{bmatrix} 0 \\ 1 \\ -\boldsymbol{B}^{-1}\boldsymbol{a}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -1 \\ -4 \end{bmatrix}$

[1] Basic Feasible Solution, *i.e.* corner

- $\bar{c}_1 < 0$ and $\bar{c}_2 < 0$ so either edge is acceptable. We'll take edge $j = 1$. Note that because $\boldsymbol{\delta}_1 \not\geq \mathbf{0}$, this edge is **unbounded**.
- **Computing $\epsilon_{\max}$.** Recall $\epsilon_{\max} = \min\left\{ \dfrac{x_i}{-(B^{-1}\boldsymbol{a}_1)_i} : i = 3, 4 \right\}$
- **Compute:** $\frac{x_3}{(\boldsymbol{B}^{-1}\boldsymbol{a}_1)_3} = \frac{3}{2}$ and $\frac{x_4}{(\boldsymbol{B}^{-1}\boldsymbol{a}_1)_4} = \frac{4}{1} = 4$

$$
\begin{aligned}
i^* &= \arg\min\left\{ \frac{x_i}{-(B^{-1}\boldsymbol{a}_1)_i} : \ -(\boldsymbol{B}^{-1}\boldsymbol{a}_1)_i < 0 \right\} \\
&= \arg\min\left\{ \frac{x_i}{-(B^{-1}\boldsymbol{a}_1)_i} : i = 3, 4 \right\} \\
&= \arg\min\left\{ \left(\frac{3}{2}\right)_3, (4)_4 \right\} = 3
\end{aligned}
$$

$$
\text{and } \epsilon_{\max} = \min\left\{ \frac{3}{2}, 4 \right\}
$$

- So: $B^{\text{new}} = \{1, 4\}$, $\boldsymbol{x}_{B^{\text{new}}} = \begin{bmatrix} 0 \\ 0 \\ 3 \\ 4 \end{bmatrix} + \frac{3}{2} \begin{bmatrix} 1 \\ 0 \\ -2 \\ -1 \end{bmatrix} = \begin{bmatrix} 3/2 \\ 0 \\ 0 \\ 5/2 \end{bmatrix}$

Non-simplex methods

- At worst, the simplex method may visit a large fraction of the corners of $\mathcal{X} = \{\boldsymbol{x} : \ \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \text{ and} \boldsymbol{x} \geq 0\}$.
- **Klee-Minty** example: simplex method visits $2^n - 1$ corners, hence simplex method has **worst case** complexity of $O(2^n)$.
- **Average Complexity**, under certain probability distribution on LP's, is however polynomial. In practice simplex method wroks reasonably fast.
- **Khachiyen's method** (aka Ellipsoid Method) has complexity $O(n^4 L)$. Not great in practice.
- **Kamarkar's method** (an Interior Point Method) has complexity $O(n^{3.5} L)$. Performs well in practice.
- Textbook (Chong & Zak) covers **Affine Scaling Method**, a simple interior point method.

Software

There exist several good packages for solving LP's. For Example:

- CVX. `http://cvxr.com/`
  - More of a framework for converting problems to LP's.
  - Works best with MATLAB.
  - Free.
- Gurobi `https://www.gurobi.com/`
  - Powerful and fast implementation of simplex method.[2]
  - Free for academic users, not free for business.
  - Works well with Python and MATLAB.

[2] I think? They don't explicitly say which method they use.